

連立型① - 2変数をもつとき -

$$\begin{cases} a_{n+1} = p a_n + q b_n \\ b_{n+1} = r a_n + s b_n \end{cases}$$

$$a_{n+1} - \alpha b_{n+1} = \beta (a_n - \alpha b_n)$$

と変形して④ 等比型へ

$$\text{例 A} \quad \begin{cases} a_{n+1} = 4a_n + 2b_n \\ b_{n+1} = a_n + 3b_n \end{cases} \quad a_1 = 3, \quad b_1 = 2$$

$$a_{n+1} - \alpha b_{n+1} = \beta (a_n - \alpha b_n) \dots \text{①}$$

とあくと

$$4a_n + 2b_n - \alpha(a_n + 3b_n) = \beta(a_n - \alpha b_n)$$

$$(4 - \alpha)a_n + (2 - 3\alpha)b_n = \beta a_n - \alpha\beta b_n$$

よって

$$\begin{cases} 4 - \alpha = \beta \\ 2 - 3\alpha = -\alpha\beta \end{cases}$$

$$\therefore (\alpha, \beta) = (-1, 5) (2, 2)$$

これらを①に代入して

$$\begin{cases} a_{n+1} + b_{n+1} = 5(a_n + b_n) \dots \text{②} \\ a_{n+1} - 2b_{n+1} = 2(a_n - 2b_n) \dots \text{③} \end{cases}$$

② より

$$\{a_n + b_n\} \text{ は } \textcircled{2} a_1 + b_1, \textcircled{2} 5 \text{ の等比数列}$$

$$\begin{aligned} a_n + b_n &= (a_1 + b_1) \cdot 5^{n-1} \\ &= 5^n \dots \text{④} \end{aligned}$$

③ より

$$\{a_n - 2b_n\} \text{ は } \textcircled{3} a_1 - 2b_1, \textcircled{2} 2 \text{ の等比数列}$$

$$\begin{aligned} a_n - 2b_n &= (a_1 - 2b_1) \cdot 2^{n-1} \\ &= -2^{n-1} \dots \text{⑤} \end{aligned}$$

④ × 2 + ⑤ より

$$\begin{aligned} 3a_n &= 2 \cdot 5^n - 2^{n-1} \\ \therefore a_n &= \frac{1}{3} (2 \cdot 5^n - 2^{n-1}) // \end{aligned}$$

④ - ⑤ より

$$\begin{aligned} 3b_n &= 5^n + 2^{n-1} \\ \therefore b_n &= \frac{1}{3} (5^n + 2^{n-1}) // \end{aligned}$$